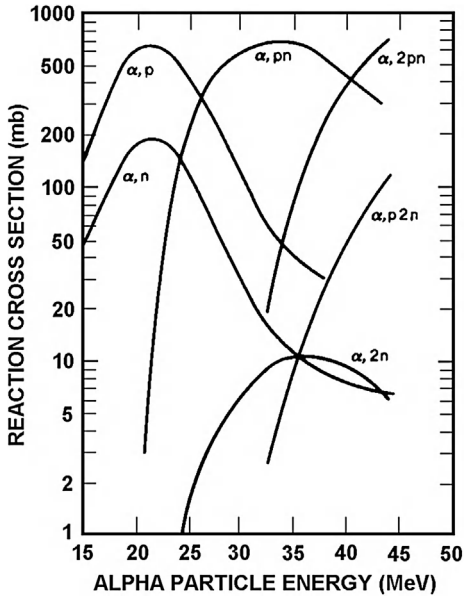


Figure 11.2 Excitation functions for reactions between ^{4}He ions and ^{54}Fe target nuclei. The kinetic energy of the projectile is in the laboratory system. (From Houck and Miller.)

11.3. RESONANCE AND TUNNELING

Experimentally, it is found that nuclear reactions sometimes occur at energies less than that required by the Coulomb barrier. This behavior is related to the wave mechanical nature of the particles involved in a nuclear reaction.

As a projectile approaches a target nucleus in a nuclear reaction, the probability that there will be overlap and hence interaction in their wave functions increases. This concept was used in §6.7.3 to explain the emission of α-particles with energies less than that required to overcome the Coulomb barrier. Such tunneling may also occur for projectiles approaching the nucleus from the outside. An example is provided by the reaction of protons with lithium (Fig. 11.3).

$$^{7}_{3}\mathrm{Li} + {}^{1}_{1}\mathrm{H} \rightarrow {}^{8}_{4}\mathrm{Be}^{*} \rightarrow \begin{cases} 2\,{}^{4}_{2}\mathrm{He} & Q = 17.4\ \mathrm{MeV} \quad (11.10a) \\ {}^{8}_{4}\mathrm{Be} + \gamma & Q = 17.2\ \mathrm{MeV} \quad (11.10b) \\ {}^{7}_{4}\mathrm{Be} + \mathrm{n} & Q = -1.64\ \mathrm{MeV} \quad (11.10c) \end{cases}$$

For this reaction the value of $E_{\mathrm{cb(min)}}$ is 1.3 MeV. However, due to tunneling the reactions begin to occur at lower proton energies. At an energy of 0.15 MeV about 0.1% of the protons penetrate the Coulomb barrier, at 0.3 MeV about 1%, and at 0.6 MeV about 20%.

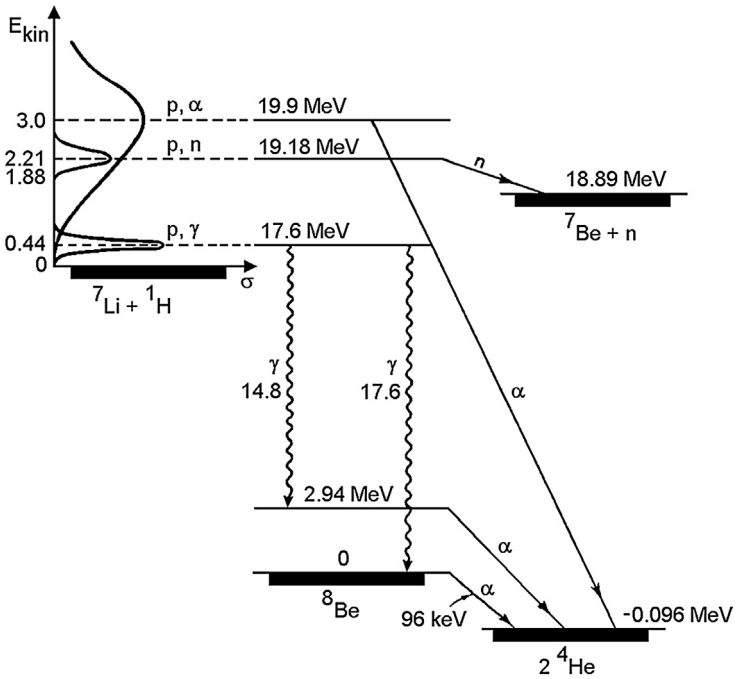


Figure 11.3 Yield curves for the reaction between protons and ^{7}Li, leading to different excited levels in ^{8}Be, followed by decay to stable end products.

The reaction cross-section is closely related to the excited energy states of the compound nucleus. Four such levels are shown for $^{8}Be^{*}$ in Figure 11.3. To the left of the figure the (p,γ), (p,n) and (p,α) partial cross-sections (excitation functions) are shown as function of the proton kinetic energy. The maximum cross-section for reaction (b) occurs at a proton kinetic energy of 0.44 MeV, which, together with the *Q*-value, 17.2 MeV, of the reaction $^{7}Li + {}^{1}H \rightarrow {}^{8}Be$, leads to an excitation energy of 17.6 MeV, which exactly matches an excited level of the same energy in $^{8}Be^{*}$. At an excitation energy of 19.18 MeV another energy level is reached in the compound nucleus leading to its decay into $^{7}Be + n$, reaction (c). The excitation energy is achieved from the release of nuclear binding energy (17.25 MeV) and from the proton kinetic energy. The amount needed is $19.18 - 17.25 = 1.93$ MeV. In order to conserve momentum, the proton must have $(8/7) \times 1.93 = 2.21$ MeV in kinetic energy. The increase in cross-section when the total excitation energy matches an excited energy level of the compound nucleus is known as a *resonance*.

This particular reaction is of interest for several reasons. It was the first nuclear reaction that was produced in a laboratory by means of artificially accelerated particles (Cockcroft and Walton 1932; cf. §16.3). Reaction (b) is still used for the production of γ-radiation (17 MeV), while reaction (c) is used as a source of mono-energetic neutrons. The energy of the neutrons from reaction (c) is a function of the proton energy and the

angle between the neutron and the incident proton beam. A necessary requirement, however, is that the threshold energy ($1.64 \times (8/7) = 1.88$ MeV) must be exceeded, the Q-value for reaction (c) being -1.64 MeV.

11.4. NEUTRON CAPTURE AND SCATTERING

Unlike charged particles, no Coulomb barrier hinders neutrons from reaching the target nucleus. This leads to generally higher reaction cross-sections for neutrons, particularly at very low energies. Moreover, since neutrons can be produced in very high fluxes in nuclear reactors, neutron-induced processes are among the more important and well studied nuclear reactions.

We have seen that the geometric cross-section of a target nucleus is in the order of 1 b, or 10^{-28} m^2. Experimentally, the cross-sections for capture of energetic ("fast") neutrons ($E_n \geq 1$ MeV) are often close to 1 b. However, for neutrons whose kinetic energy is in the $1-100$ eV region, some nuclei show very large cross-sections – as high as 10^5 b. Such values can be explained as being due to neutron capture where the compound nucleus is excited exactly to one of its discrete energy levels (resonance capture). This does not mean that the nucleus is larger than its calculated geometric cross-section but that the interaction probability is very large in such cases – greater than the calculation of πr^2 would indicate. This can be understood as a case where sometimes reaction occur between the tails of the overlapping wave functions even when the overlap is small.

For low energy ("slow") neutrons ($E_n \ll 1$ MeV) the cross-section is also larger than the πr^2 value, and decreases as the velocity increases; this relation, $\sigma \propto v^{-1}$, is shown for boron in Figure 11.4. The relationship between the cross-section and the neutron velocity can be understood in wave mechanical terms since the wavelength associated with the neutron increases with a decrease in velocity. According to the matter-wave hypothesis (§2.3) the wavelength associated with a moving particle is $\lambda = \hbar/(mv)$ which can be written:

$$\lambda = \hbar(2mE_{kin})^{-1/2} = 0.286 \times 10^{-8}(mE_{kin})^{-1/2} \tag{11.11}$$

While the wavelength of a slow neutron is about 0.1 nm, that of a fast neutron is less than 1/1000 of that. Since the reaction probability increases with increasing particle wavelength (lager wave function overlap), a slow-moving particle has a higher probability of reaction than a faster one of the same kind.

In effect, the wave properties make neutrons appear much larger than their geometric size and increases the probability of interaction with the nucleus. From (11.9) it follows that

$$\sigma_{capt} \propto \lambda_n \propto E_n^{-1/2} \propto v_n^{-1} \tag{11.12}$$

where λ_n, E_n, and v_n are the wavelength, kinetic energy, and velocity, respectively, of the neutron. This relation is known as the $1/v$ law. This law is valid only where no resonance